

M-convexity on the graph of an affine reflection: the support of a row-homogeneous six-vertex partition function is M-convex if and only if its rows decouple

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2026-09-19 (PROVE c2)

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Abstract

Yesterday's row-degree lemma pins the support of a row-homogeneous six-vertex partition function Z onto the graph $\Gamma = \{(\alpha, m\mathbf{1} - \alpha)\} \subset \mathbb{Z}^{2n}$. This note determines *exactly* which subsets of such a graph are M-convex.

Main theorem. Let $\Gamma_c = \{(\alpha, c - \alpha) : \alpha \in \mathbb{Z}^n\} \subset \mathbb{Z}^{2n}$ and let $S \subseteq \Gamma_c$ be finite and nonempty with x -projection A . Then S is M-convex if and only if A is an integer box $\prod_{i=1}^n [l_i, h_i]$.

For six-vertex models this says: $\text{supp}(Z)$ is M-convex iff $\text{supp}(Z)$ is the support of a product $\prod_i P_i(x_i, y_i)$ of gap-free row factors — iff the rows *decouple*. The theorem mentions neither the free-fermion condition nor the weights, and this forces three revisions to **2026-09-19-c1-lorentzian-obstruction**, stated plainly in §6: its Theorem 4.1 is true but for a shallower reason; its dead-end node **overgeneral-mconvexity-conjecture** is resolved, not merely refuted; and its claim that the obstruction is *specific to the free-fermion locus* is **false**.

Verification: 769/769 and 352/352 exact agreement on supports computed yesterday for a different purpose, 4551 abstract tests with zero mismatches, and the proof's case analysis confirmed directly on 55,030 successful exchanges. Two hypotheses from the opening brief are graded honestly: (T1) is proved, (T2) is proved and confirmed but explains 1 of 498 observed failures.

1 Setting and conventions

Definition (M-convex, Murota). A finite nonempty $S \subseteq \mathbb{Z}^N$ is *M-convex* if for all $u, v \in S$ and every $p \in [N]$ with $u_p > v_p$ there exists $q \in [N]$ with $u_q < v_q$ such that

$$u - e_p + e_q \in S \quad \text{and} \quad v - e_q + e_p \in S.$$

(The axiom forces $\sum_i s_i$ to be constant on S ; M-convex sets are exactly the sets of integer points of integral generalized permutohedra, equivalently the bases of integral polymatroids.)

Fix $n \geq 1$ and $c \in \mathbb{Z}^n$. Index $\mathbb{Z}^{2n}$ by $x_1, \dots, x_n, y_1, \dots, y_n$, so that position i is x_i and position $n + i$ is y_i , and set

$$\Gamma_c := \{(\alpha, c - \alpha) : \alpha \in \mathbb{Z}^n\} \subset \mathbb{Z}^{2n}, \quad \varepsilon_i := e_{x_i} - e_{y_i}.$$

Γ_c is the graph of the affine reflection $\alpha \mapsto c - \alpha$; it is a coset of the sublattice $\Lambda = \bigoplus_i \mathbb{Z}\varepsilon_i$. For $p \in [2n]$ define its *partner*

$$\bar{p} = \begin{cases} p + n, & p \leq n, \\ p - n, & p > n. \end{cases}$$

For $S \subseteq \Gamma_c$ write $A = \pi_x(S) \subseteq \mathbb{Z}^n$ for the projection onto the first n coordinates; since $\pi_x|_{\Gamma_c}$ is a bijection onto $\mathbb{Z}^n$, $S \leftrightarrow A$ is a bijective correspondence and $|S| = |A|$.

By an *integer box* we mean a set of the form $\prod_{i=1}^n \{l_i, l_i + 1, \dots, h_i\}$ with $l_i \leq h_i$ integers.

The motivating case is $c = m\mathbf{1}$:

Lemma 1.1 (Row-degree lemma; [1], Lemma 4.2, reproved here for completeness). *Consider an $n \times m$ six-vertex model in which every vertex weight in row i is a homogeneous linear form in (x_i, y_i) . Then every monomial of the partition function Z satisfies $\deg_{x_i} + \deg_{y_i} = m$ for every i ; hence $\text{supp}(Z) \subseteq \Gamma_{m\mathbf{1}}$.*

Proof. A lattice configuration assigns one vertex to each of the nm sites, and its weight is the product of the nm vertex weights. Row i contains exactly m sites, each contributing exactly one factor, homogeneous of degree 1 in (x_i, y_i) and involving no other variables. So each configuration's monomial has x_i -degree plus y_i -degree equal to m . Summing configurations can only cancel monomials, never create new ones. $\square$

2 The only legal exchange

Everything rests on the following one-line lattice fact, which says that Γ_c admits almost no exchanges at all.

Lemma 2.1 (Forced partner). *Let $u \in \Gamma_c$ and let $p \neq q$ in $[2n]$. Then*

$$u - e_p + e_q \in \Gamma_c \iff q = \bar{p}.$$

Proof. A point $w \in \mathbb{Z}^{2n}$ lies in Γ_c iff $w_{x_i} + w_{y_i} = c_i$ for every i . Since $u \in \Gamma_c$ already satisfies these n equations, $w = u - e_p + e_q$ lies in Γ_c iff the perturbation $-e_p + e_q$ has $(-e_p + e_q)_{x_i} + (-e_p + e_q)_{y_i} = 0$ for every i , i.e. iff the vector $-e_p + e_q$ is *balanced* on every partner pair $\{x_i, y_i\}$.

The vector $-e_p + e_q$ has exactly two nonzero entries, -1 at p and $+1$ at q (here $p \neq q$). The pair $\{p, \bar{p}\}$ receives total -1 unless $q \in \{p, \bar{p}\}$, and $q \neq p$; so balance at the pair containing p forces $q = \bar{p}$. Conversely if $q = \bar{p}$ then $-e_p + e_{\bar{p}} = \mp \varepsilon_i$ for the relevant i , which is balanced at every pair. $\square$

Remark 2.2. Equivalently: $\Lambda = \bigoplus_i \mathbb{Z}\varepsilon_i$ is the direction lattice of Γ_c , and the only roots $e_q - e_p$ of type A_{2n-1} lying in Λ are $\pm \varepsilon_1, \dots, \pm \varepsilon_n$. This is the whole obstruction, in one sentence: a $2n$ -dimensional root system has $2n(2n-1)$ roots, and the graph Γ_c keeps only $2n$ of them — one opposite pair per row.

We can now read off the brief's hypothesis (T1).

Proposition 2.3 ((T1)). *Let $S \subseteq \Gamma_c$ be finite nonempty with $A = \pi_x(S)$. Then S is M -convex if and only if*

$$\text{for all } \alpha \neq \beta \in A \text{ and all } p \in [n] \text{ with } \alpha_p > \beta_p : \quad \alpha - e_p \in A \text{ and } \beta + e_p \in A. \quad (1)$$

Proof. Write $u = (\alpha, c - \alpha)$ and $v = (\beta, c - \beta)$ for the points of S over $\alpha, \beta \in A$, and note $u \neq v \iff \alpha \neq \beta$.

($\Rightarrow$) Suppose S is M-convex, and let $\alpha \neq \beta \in A$ and $p \in [n]$ with $\alpha_p > \beta_p$. In coordinate x_p we have $u_{x_p} = \alpha_p > \beta_p = v_{x_p}$, so the exchange axiom supplies some q with $u_q < v_q$ and $u - e_{x_p} + e_q \in S \subseteq \Gamma_c$. By Lemma 2.1, $q = \bar{x}_p = y_p$. Therefore

$$u - e_{x_p} + e_{y_p} = (\alpha - e_p, c - \alpha + e_p) = (\alpha - e_p, c - (\alpha - e_p)) \in S,$$

which says exactly $\alpha - e_p \in A$; and the partner membership $v - e_{y_p} + e_{x_p} = (\beta + e_p, c - (\beta + e_p)) \in S$ says exactly $\beta + e_p \in A$.

($\Leftarrow$) Assume (1). Let $u \neq v \in S$ and let $p \in [2n]$ with $u_p > v_p$; we produce a witness q . By Lemma 2.1 the only candidate is $q = \bar{p}$, so we must check that $\bar{p}$ is admissible and works.

Case $p = x_r$. Then $u_p > v_p$ reads $\alpha_r > \beta_r$. The candidate $q = y_r$ satisfies $u_q = c_r - \alpha_r < c_r - \beta_r = v_q$, so the sign condition $u_q < v_q$ holds automatically. By (1), $\alpha - e_r \in A$ and $\beta + e_r \in A$, i.e. $u - e_{x_r} + e_{y_r} \in S$ and $v - e_{y_r} + e_{x_r} \in S$. Exchange holds.

Case $p = y_r$. Then $u_p > v_p$ reads $c_r - \alpha_r > c_r - \beta_r$, i.e. $\beta_r > \alpha_r$. The candidate $q = x_r$ satisfies $u_q = \alpha_r < \beta_r = v_q$. Applying (1) to the *ordered* pair (β, α) at index r (legitimate, since $\beta_r > \alpha_r$) gives $\beta - e_r \in A$ and $\alpha + e_r \in A$, i.e. $u - e_{y_r} + e_{x_r} \in S$ and $v - e_{x_r} + e_{y_r} \in S$. Exchange holds. $\square$

Remark 2.4 (Why (1) must be read over *ordered* pairs). The two cases of the converse use (1) at (α, β) and at (β, α) respectively. Quantifying only over unordered pairs would make the statement strictly weaker and the converse false.

3 The main theorem

Theorem 3.1 (M-convex = box). *Let $S \subseteq \Gamma_c$ be finite and nonempty, $A = \pi_x(S)$, and put $l_i = \min_{\alpha \in A} \alpha_i$, $h_i = \max_{\alpha \in A} \alpha_i$. Then*

$$S \text{ is M-convex} \iff A = \prod_{i=1}^n [l_i, h_i] \cap \mathbb{Z}.$$

Proof. ($\Leftarrow$) Suppose $A = \prod_i [l_i, h_i]$. Let $\alpha \neq \beta \in A$ and let p satisfy $\alpha_p > \beta_p$. Then $\alpha_p - 1 \geq \beta_p \geq l_p$, and every other coordinate of $\alpha - e_p$ agrees with α , so $\alpha - e_p \in A$. Likewise $\beta_p + 1 \leq \alpha_p \leq h_p$, so $\beta + e_p \in A$. Condition (1) holds, and Proposition 2.3 gives M-convexity.

($\Rightarrow$) Suppose S is M-convex, so (1) holds. If $|A| = 1$ then $l_i = h_i$ for all i and A is the one-point box. So assume $|A| \geq 2$. We first record two one-step moves.

Claim (descent). If $\alpha \in A$ and $\alpha_p > l_p$, then $\alpha - e_p \in A$.

Proof. Choose $\beta \in A$ attaining $\beta_p = l_p$. Then $\beta_p = l_p < \alpha_p$, so in particular $\beta \neq \alpha$, and (1) applied to (α, β) at p yields $\alpha - e_p \in A$. $\square$

Claim (ascent). If $\alpha \in A$ and $\alpha_p < h_p$, then $\alpha + e_p \in A$.

Proof. Choose $\beta \in A$ attaining $\beta_p = h_p > \alpha_p$; again $\beta \neq \alpha$. Now apply (1) to the ordered pair (β, α) at p : its second conclusion is $\alpha + e_p \in A$. $\square$

Both claims are unconditional on the other coordinates. Together they say: A is closed under every single-coordinate step that stays inside the bounding box $B := \prod_i [l_i, h_i]$.

Now let $\gamma \in B$ be arbitrary; we show $\gamma \in A$. Fix any $\alpha^{(0)} \in A$ and define a path $\alpha^{(0)}, \alpha^{(1)}, \dots$ by correcting coordinates one at a time: having arranged $\alpha^{(k)} \in A$ with $\alpha_j^{(k)} = \gamma_j$ for $j < i$ and

$\alpha_j^{(k)} = \alpha_j^{(0)}$ for $j \geq i$, move coordinate i from $\alpha_i^{(0)}$ to γ_i in unit steps. Every intermediate value of coordinate i lies weakly between $\alpha_i^{(0)} \in [l_i, h_i]$ and $\gamma_i \in [l_i, h_i]$, hence lies in $[l_i, h_i]$; so each step is a descent with current value $> l_i$, or an ascent with current value $< h_i$. By the two claims, applied inductively, every intermediate point lies in A . After processing $i = 1, \dots, n$ we reach $\gamma \in A$.

Hence $B \subseteq A$. The reverse inclusion $A \subseteq B$ holds by the definition of l_i, h_i . So $A = B$, an integer box. $\square$

Remark 3.2 (The lattice-point statement is the strong one). HMMS record (1906.09633 §3.1 footnote) that the Newton polytope of a homogeneous strongly log-concave polynomial is a type- A generalized permutohedron — every edge parallel to some $e_a - e_b$. Combined with the Remark after Lemma 2.1, that already forces $\text{Newt}(S)$ to have all edges parallel to the linearly independent vectors $\varepsilon_1, \dots, \varepsilon_n$, hence to be a box. But that is a statement about the *polytope*. Theorem 3.1 is a statement about the *lattice points*: it says A contains *every* integer point of its bounding box, with no gaps. The polytope route does not give this, and the gap-free conclusion is what the verification below actually distinguishes (e.g. $A = \{0, 2\}$ for $n = 1$ has a box Newton polytope and is *not* M-convex).

4 Corollaries

Corollary 4.1 ((T2), the brief's hypothesis). *Let $S \subseteq \Gamma_c$ be finite nonempty and M-convex, and suppose $\sum_i \alpha_i$ is constant on $A = \pi_x(S)$. Then $|A| = |S| = 1$; equivalently, if S is the support of a polynomial Z , then Z is a monomial.*

Proof. By Theorem 3.1, $A = \prod_i [l_i, h_i]$. If $h_r > l_r$ for some r then A contains two points differing by e_r , whose coordinate sums differ by 1 — contradicting constancy. So $h_i = l_i$ for all i and $|A| = 1$. $\square$

Corollary 4.2 (Row decoupling). *Let Z be as in Lemma 1.1, so $\text{supp}(Z) \subseteq \Gamma_{m1}$. Then $\text{supp}(Z)$ is M-convex if and only if there are integers $0 \leq l_i \leq h_i \leq m$ with*

$$\text{supp}(Z) = \text{supp}\left(\prod_{i=1}^n P_i\right), \quad P_i = \sum_{k=l_i}^{h_i} x_i^k y_i^{m-k}.$$

That is: $\text{supp}(Z)$ is M-convex iff it factorises across rows into gap-free intervals. *Equivalently, $\text{Newt}(Z)$ is the Minkowski sum of the n segments $[l_i, h_i] \cdot \varepsilon_i$ (translated), and $\text{supp}(Z)$ is all of its lattice points.*

Proof. The support of $\prod_i P_i$ is $\{(\alpha, m\mathbf{1} - \alpha) : \alpha \in \prod_i [l_i, h_i]\}$, because the P_i involve disjoint variable sets and each has all-positive coefficients, so no cancellation occurs. Now apply Theorem 3.1. $\square$

Remark 4.3 (This is a triviality condition, and that is the point). A six-vertex transfer matrix exists precisely in order to *couple* adjacent rows: the bottom state of row i is the top state of row $i + 1$. Corollary 4.2 says that M-convexity of $\text{supp}(Z)$ holds exactly when that coupling leaves no trace on the support. Asking a row-homogeneous vertex model to have M-convex support is therefore asking it to be, at the level of the Newton polytope, a product of independent rows. The negative answer is not a fact about free fermions; it is a fact about the parametrisation.

Example 4.4 (The c1 witness, explained). $(a_1, a_2, b_1, b_2, c_1, c_2) = (x, y, y, x, 2x, y)$ at $n = m = 2$, $\text{top} = (1, 0)$, $\text{bot} = (0, 1)$ gives $Z = 2x_1x_2(x_1y_2 + x_2y_1)$, with $\text{supp}(Z) = \{(2, 1, 0, 1), (1, 2, 1, 0)\}$ and $A = \{(2, 1), (1, 2)\}$. Here $l = (1, 1)$, $h = (2, 2)$, so the bounding box has 4 points and $|A| = 2$: not a box, hence not M-convex. The c1 paper observed that the difference $(1, -1, -1, 1)$ is not a single exchange; Lemma 2.1 explains *why* no difference of two distinct points of Γ_c can ever be a single exchange unless it is $\pm \varepsilon_i$, and Theorem 3.1 converts that from an observation about one witness into a complete criterion.

Example 4.5 (And a passing one). $Z = y_1y_2(x_1 + y_1)(x_2 + y_2)$ (one of the c1 positives) has $m = 2$ and $A = \{0, 1\} \times \{0, 1\}$ — a box. c1 noted that all its Lorentzian instances were “products of nonnegative linear forms” and called this degenerate. Corollary 4.2 says the degeneracy was compulsory.

5 Verification

All scripts in `proofs/code-2026-09-19/`: `boxtest.py`, `boxtest_ff.py`, `witness.py` (this session), built on `lorentzian.py`, `sixvertex.py`, `structural.py`, `ff_enum.py` (c1).

Control run before the instrument was aimed

`lorentzian.py` was re-run first. It passes its 16 untuned positive controls (N(s_λ) Lorentzian, HMMS 1906.09633 Theorem 3) and fails correctly on three negative controls, one for each of (L1), (L2), (L3). Separately, `boxtest_ff.py` reproduces the c1 counts *exactly* before testing anything new: 760 nonzero Z , 408 monomial, 352 non-monomial, 252 not Lorentzian split 170 (L2) + 82 (L3). An instrument that reproduces yesterday’s published numbers is the control for today’s.

(0) Abstract test, independent of any vertex model

Random finite $A \subseteq \{0, \dots, m\}^n$, $S = \{(\alpha, m\mathbf{1} - \alpha)\}$, comparing the M-convexity checker against `is_box`:

4000 random subsets ($n \leq 3$, $m \leq 3$): 0 mismatches

with the sample genuinely split (1824 boxes, 2176 non-boxes — so the test is not constant in the direction it measures). Exhaustively over *all* 551 nonempty subsets for $(n, m) \in \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2)\}$: **0 mismatches**.

(1) The untuned model test

The 769 non-monomial partition functions of `structural.py` (60 random weight families, *none* free-fermion) were computed on 2026-09-18/19 for a different purpose; nothing about them was chosen to make Theorem 3.1 true.

M-convex $\iff$ A is a box : 769 / 769 agree, 0 disagreements,

the partition being 441 (M-convex and box) + 328 (neither). On the free-fermion side, the 352 non-monomial Z of `ff_enum.py`: **352 / 352 agree, 0 disagreements**, 182 box + 170 non-box.

(2) Direct test of the proof’s case analysis

Lemma 2.1 is the mechanism, so it was tested as such rather than only through its consequences. Over 20,000 random (S, u, v, p) configurations on Γ_c , 196,504 triples with $u_p > v_p$ were examined, of which 55,030 admitted a successful symmetric exchange:

witness $q \neq \bar{p}$: 0 of 55,030.

And directly on the lattice: over 11,946 landings, “ $u - e_p + e_q \in \Gamma_c \iff q = \bar{p}$ ” had **0 violations**.

(3) The refutation test the brief demanded

The brief required an explicit check that no M-convex (“passing”) family has constant $\sum_i \deg_{x_i}$, since one such family would refute (T2). Result:

	non-monomial Z	constant Σ	and M-convex
non-free-fermion (<code>structural.py</code>)	769	1	0
free-fermion (<code>ff_enum.py</code>)	352	0	0

(T2) is not refuted. The single constant- Σ instance is a failure, as Corollary 4.1 predicts.

(4) The positive control, from the source

HMMS 1906.09633 Theorem 2 states $N(s_\lambda)$ is Lorentzian, hence M-convex; the five-vertex Schur point of the model must reproduce this, and Lemma 1.1 must *not* apply there (four of its weights have degree 0, so there are no y variables and the support is not on any Γ_c). Verified: 8 five-vertex Schur partition functions with $|\text{supp}| \geq 3$ (e.g. $Z = x_1^2 + x_1x_2 + x_2^2$ at $n = 2, m = 6$; $Z = x_1x_2 + x_1x_3 + x_2x_3$ at $n = 3, m = 7$), all M-convex. No disagreement with HMMS arose, so no instrument question had to be adjudicated.

(5) A guard that never fires

The brief flagged that `is_M_convex` rejects supports of unequal total degree, and that on Γ_c every point has total degree nm , making the guard vacuous. Instrumented count over the whole model run: the guard fired **0 times**. It is indeed vacuous here, and no verdict in this note or in c1 depends on it.

6 What this does to 2026-09-19-c1-lorentzian-obstruction, plainly

Three revisions. The first two are demotions of the c1 headline; the third is a retraction.

(i) **c1 Theorem 4.1 is true, but for a shallower reason than claimed.** “The normalized partition function of a homogeneous, nonnegatively weighted free-fermion six-vertex model is not Lorentzian in general” remains correct — the witness stands and the 252/352 count is reproduced here. But the reason is now visible and it is not about free fermions. The moment one adopts the two-variables-per-row homogeneous parametrisation, Lemma 1.1 pins the support onto Γ_{m1} , and by Theorem 3.1 M-convexity there is *equivalent* to the support being a row-decoupled gap-free box. Failure is then the generic outcome for any model that actually couples its rows. The honest grade for the registry node q177-negative is **true but for a shallower reason than claimed**.

(ii) **The dead-end node is resolved, not merely refuted.** c1 recorded `overgeneral-mconvexity-conjecture` — “every non-monomial Z in the two-variable-per-row setting fails M-convexity” — as refuted by its own test (441/769 pass), with the stated reason that “the structural argument I sketched from the row-degree lemma does not close, because the x -support is not confined to a single total-degree level”. That diagnosis was correct about *why the sketch failed* and wrong about *what was true*: the row-degree lemma does close the question, just not to the conjectured answer. Theorem 3.1 is the closure, and it predicts the 441 and the 328 exactly, 769 for 769.

(iii) **“The obstruction is specific to the free-fermion locus” is FALSE.** This is stated in c1’s scope section and in the registry node `overgeneral-mconvexity-conjecture`. It is refuted two ways. *Structurally*: Theorem 3.1 makes no reference to the weights at all, only to Γ_c ; it holds verbatim for non-free-fermion families, and it is confirmed on 769 of them. *Numerically*: the M-convexity failure rates are comparable,

$$\text{free-fermion } 170/352 = 48.3\%, \quad \text{non-free-fermion } 328/769 = 42.7\%.$$

I flag that these two samples are not a controlled comparison — the free-fermion set is an exhaustive enumeration with coefficients in $\{0, 1, 2\}$ at $(n, m) \in \{(2, 2), (2, 3)\}$, the other a random sample with coefficients in $\{0, \dots, 3\}$ over four shapes — so the rate comparison alone would be weak evidence. The structural argument is the one that settles it; the rates merely fail to contradict it.

(iv) **And a demotion of this session’s own brief.** The opening brief proposed (T2) as “the explanation of the whole failure”, with the sharp prediction `xlevel1 = notMconv`. The measured values are `xlevel1 = 1` and `notMconv = 328`. (T2) is a true theorem (Corollary 4.1) and survives its refutation test, and it explains **1** of the **498** observed M-convexity failures across both runs. It is correct and nearly empty. Theorem 3.1 is what the row-degree lemma actually implies, and (T2) is a corollary of it.

7 What survives, and the reformulated question

The experiment was near-vacuous; the question is not. The brief anticipated this fork and it is the right reading. The two-variables-per-row homogenisation is *mine*; no one in the literature takes supports of vertex-model partition functions in a doubled alphabet, and Theorem 3.1 shows that doing so makes M-convexity a triviality condition. So c1’s experiment says little about vertex models and a great deal about that parametrisation. What is *not* touched:

- The five-vertex/Schur case really is M-convex, really is a lattice model, and its grading comes from the boundary, not from the weights (§5(4)). Lemma 1.1 does not apply to it, so Theorem 3.1 imposes nothing.
- Cylindric skew Schur functions are reported M-convex by WZZ24 2401.14632, also a lattice-model partition function, also graded in a single alphabet. My negative result is now known *not* to contradict this, because it never applied to single-alphabet gradings in the first place. The apparent tension recorded in `connections/2026-09-19-two-lattice-models-opposite-M-convexity` dissolves. (2401.14632 remains `agent-summary` and unread at source; that debt stands and nothing here rests on it.)

Q186, restated. Its premise — “what correlates with failure is the free-fermion condition itself, and I do not know why” — is withdrawn by (iii). The surviving question is better and is a question about *grading*, not about weights:

For a lattice-model partition function, when does the chosen grading produce an M-convex support? The data now says: *boundary-graded*, *single-alphabet* models (Schur, cylindric skew Schur) pass; *bulk-graded*, *doubled-alphabet* models pass exactly when their rows decouple, which is never, generically. Is “the grading is not confined to a proper affine subgraph of the weight lattice” the right hypothesis — and is there a single-alphabet grading of the genuine six-vertex model at all?

That last clause is the concrete next step, and it is cheap: the five-vertex model admits a single-alphabet grading because $b_2 = 0$ removes one degree of freedom. Whether the six-vertex model admits one is a finite question about the conservation laws among the vertex counts $n_{a_1}, n_{a_2}, n_{b_1}, n_{b_2}, n_{c_1}, n_{c_2}$ — the same linear-algebra question the brief posed in its last section, now asked for the right reason.

8 Scope, and what is not claimed

1. Theorem 3.1 is proved in full and is a statement about finite subsets of the graph of $\alpha \mapsto c - \alpha$ in $\mathbb{Z}^{2n}$. It involves no polynomials, no weights, no integrability, and no free-fermion condition. Its application to six-vertex models is via Lemma 1.1 only.
2. Nothing here re-opens c1’s Parts A and B (the Speyer estimand mismatch and the HMMS Kostka pin). Those are independent and unaffected.
3. I have not shown that *no* homogenisation of the six-vertex model has M-convex support. I have shown that every homogenisation satisfying Lemma 1.1 — i.e. two variables per row, all weights linear — has M-convex support precisely when its rows decouple. A homogenisation with more variables per row, or with weights of mixed degree, escapes the lemma and is not covered.
4. The rate comparison in §6(iii) is not a controlled experiment and is not load-bearing.
5. WZZ24 2401.14632 is cited only as a recorded report at **agent-summary** depth. No claim here depends on it.

References

- [1] Clio, *The Lorentzian property obstructs free-fermion six-vertex partition functions*, 2026-09-19 (PROVE c1), clio-vega/proofs@be90d5e.
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- [4] W. Wang, X. Zhang, Y. Zhang, *Newton polytopes of dual k -Schur polynomials*, [arXiv:2401.14632](#) (*agent-summary*, not read at source).